Network Devices

Connectors, Repeaters, Hubs, Bridges, Switches, Routers, NIC's

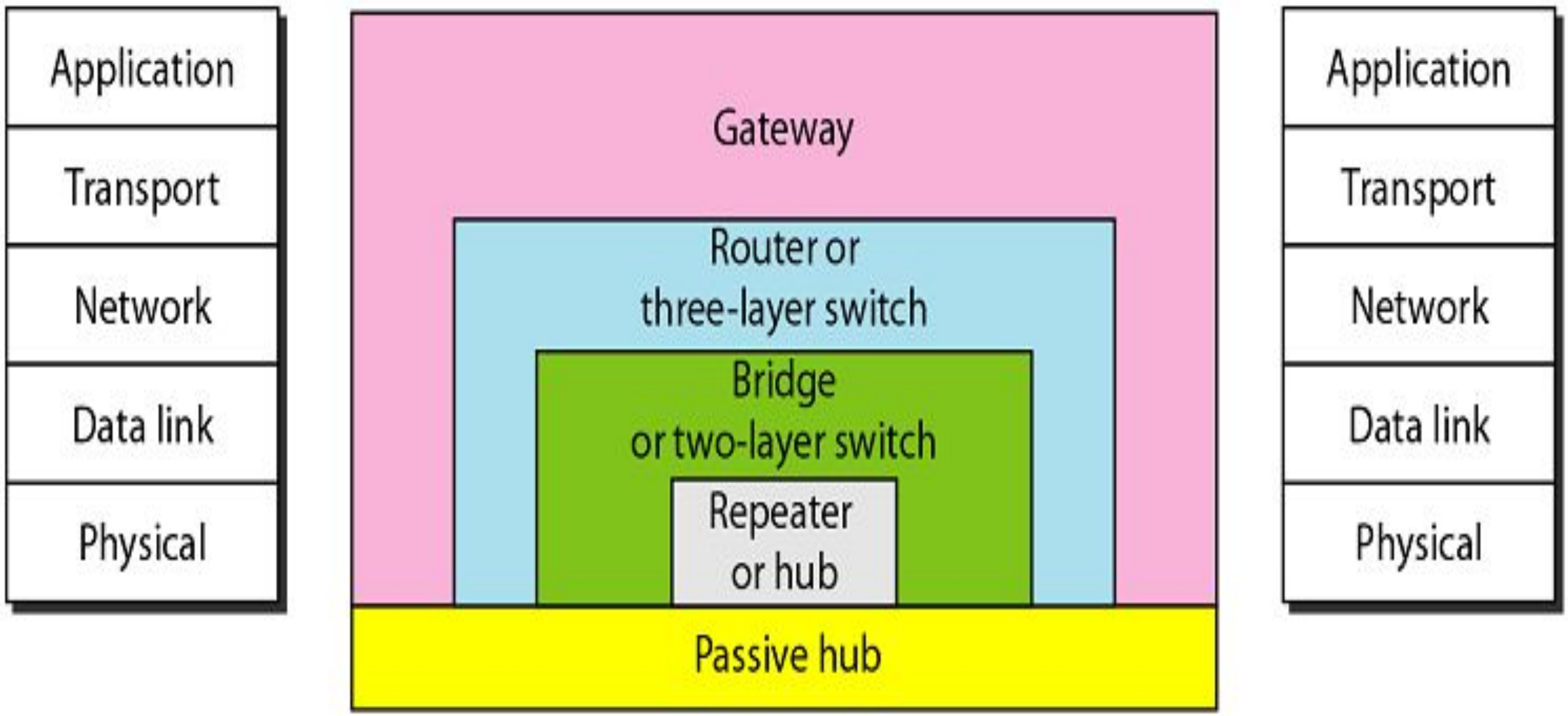
Network Devices

- Network is interconnection of devices.
- For these connection we need to use the connecting devices.
- Also called as **Network Control Devices**.

The purpose

- Allow a **greater number of nodes to be connected** to the network.
- **Extend the distance** over which a network can extend.
- **Localize traffic** on the network.
- Can **merge existing networks**.
- **Isolate network problems** so that they can be diagnosed more easily.

Devices and the layers at which they operate



Five categories of connecting devices

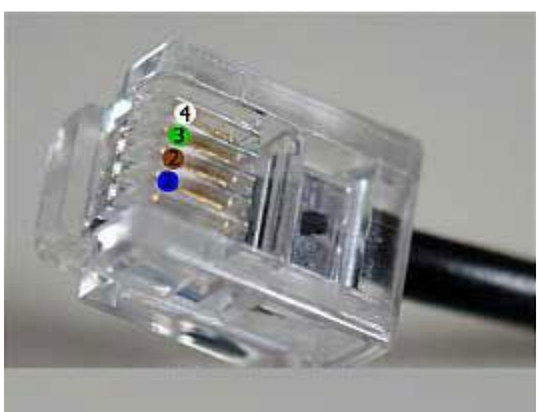
Connectors

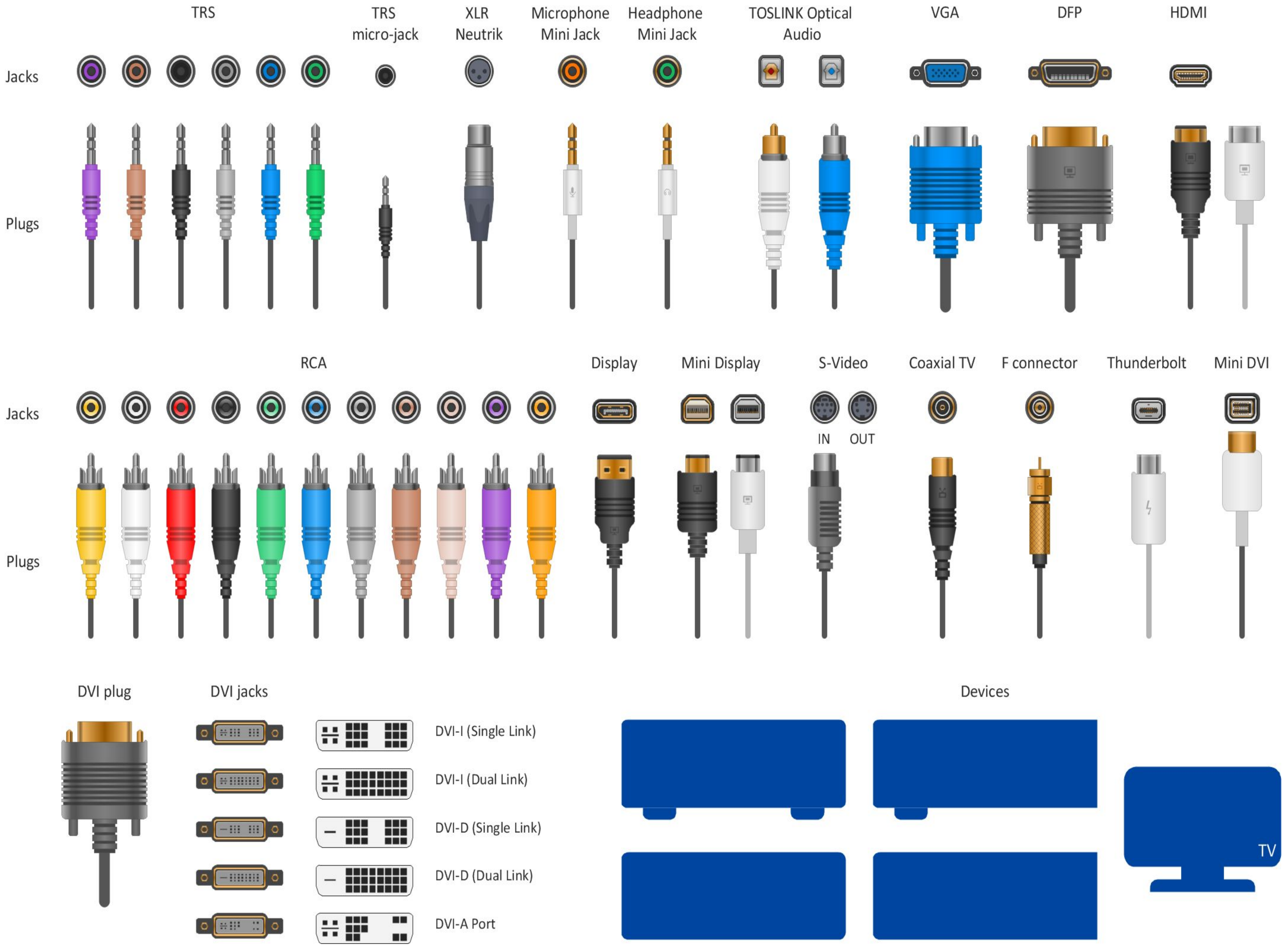
- To connect **cable between two computers**.
- Connectors are of different type such as –
  - **Twisted Pair cable**
  - **Co-axial Cable**
  - **Fibre optic cable.**
- Connectors are type such as-
  - **Jacks**
  - **Plugs**
  - **Sockets and ports**

Connectors

Example:

- RS232 and V35 for serial interface
- RJ45 and BNC connectors for Ethernet.
- SC or ST connectors for fibre optic



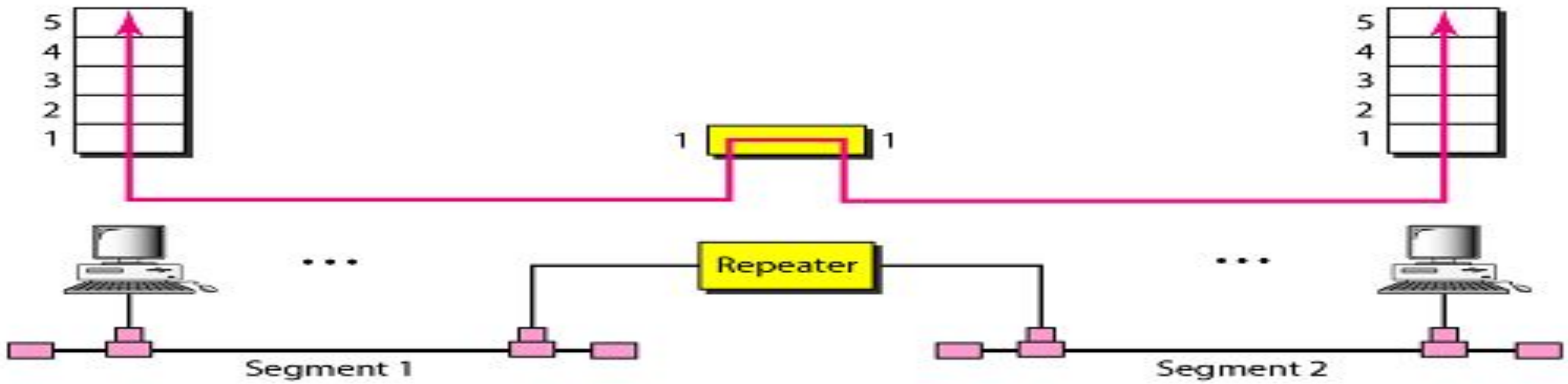

TRS
TRS micro-jack
XLR Neutrik
Microphone Mini Jack
Headphone Mini Jack
TOSLINK Optical Audio
VGA
DFP
HDMI
Jacks
Plugs
RCA
Display
Mini Display
S-Video
Coaxial TV
F connector
Thunderbolt
Mini DVI
Jacks
IN
OUT
Plugs
DVI plug
DVI jacks
DVI-I (Single Link)
DVI-I (Dual Link)
DVI-D (Single Link)
DVI-D (Dual Link)
DVI-A Port
Devices
TV

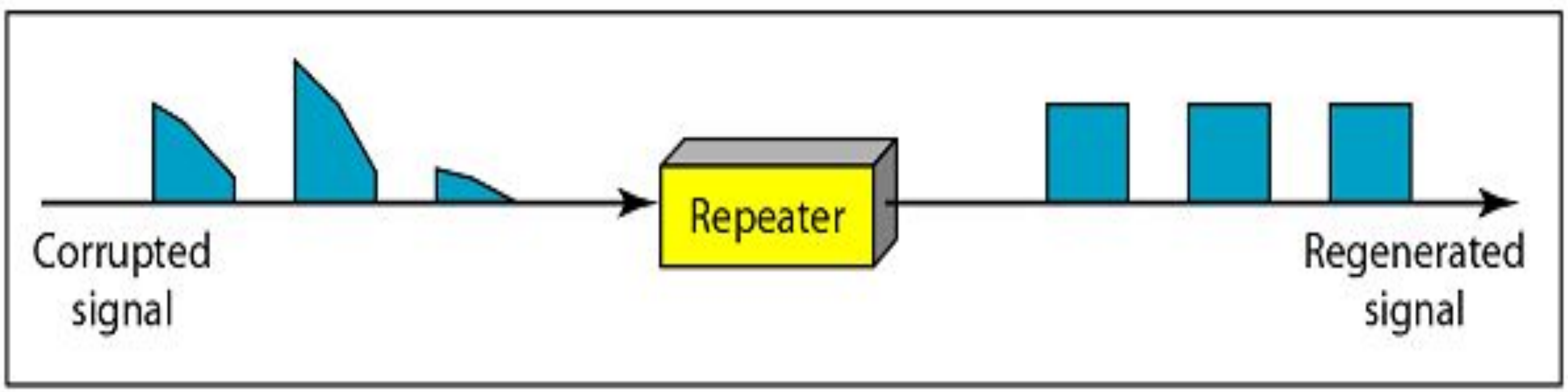
Repeaters

- **Signal attenuation or signal loss** – signal degrades over distance
- Repeaters **clean, amplify, and resend** signals that are **weakened by long cable length**.
- Built-in to hubs or switches
- A repeater operates only at the **PHYSICAL** layer.
- It connects **two segments of the same network.**
- **Single port**, **multi-port** repeaters.

Repeaters

Repeater connecting two segments of a LAN

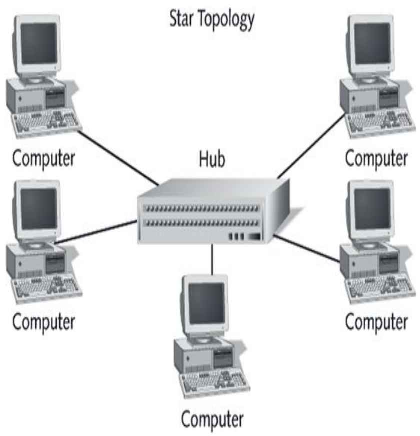




Function of a repeater

HUB

- A hub is used as a **central device**.
- Connects the computers in **star topology**.
- ***Hubs*** are simple devices that **direct data packets to all devices connected to the hub**.
- Hubs regenerate and retime network signals
- **hubs work at the OSI physical layer**
- They **cannot filter** network traffic.
- They cannot determine best path
- They are really **multi-port repeaters**



Types of Hub

1. **Passive hub**
   - is **just a connector** - connects the wires coming from different branches.
   - The signal pass through a passive hub **without regeneration or amplification**. (distance 300 feet)
2. **Active hubs or Multiport repeaters-**
   - They **regenerate or amplify** the signal before they are retransmitted (distance -2000 feet).
3. **Intelligent Hub**
   - **Regenerate** the signals and perform **network management** and **intelligent path selection**.

Bridges

- Operates in both the **PHYSICAL** and the **data link layer**.
- As a PHYSICAL layer device, it **regenerates** the signal it receives.
- As a **data link layer** device, the bridge can **check the PHYSICAL/MAC addresses** (source and destination) contained in the frame.

Bridge

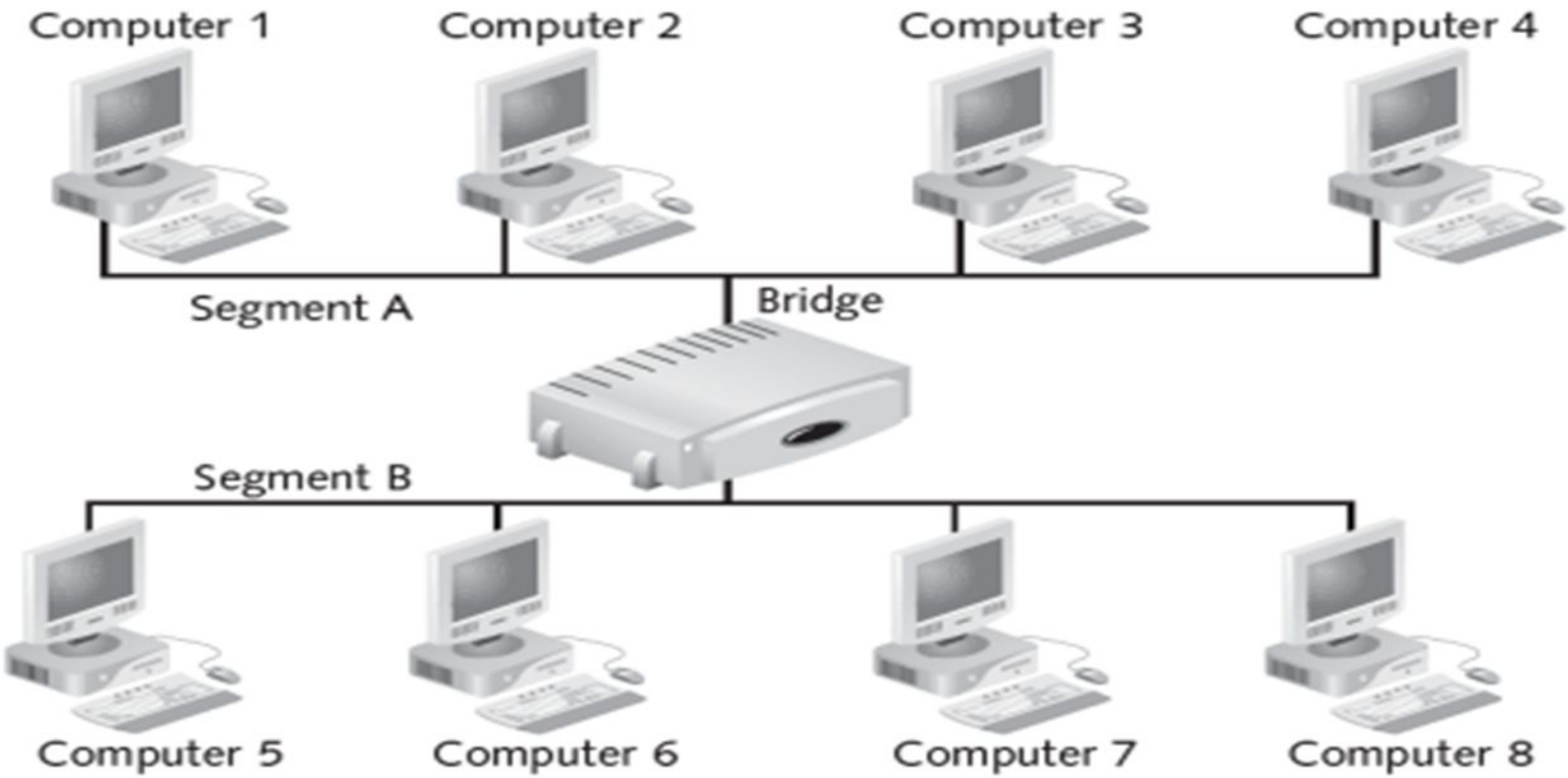
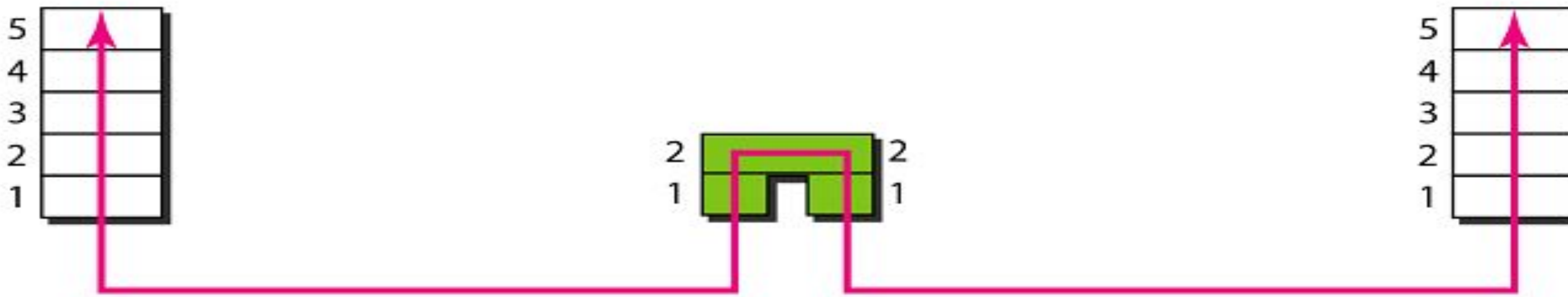


Figure 2-6 Bridge

Bridges

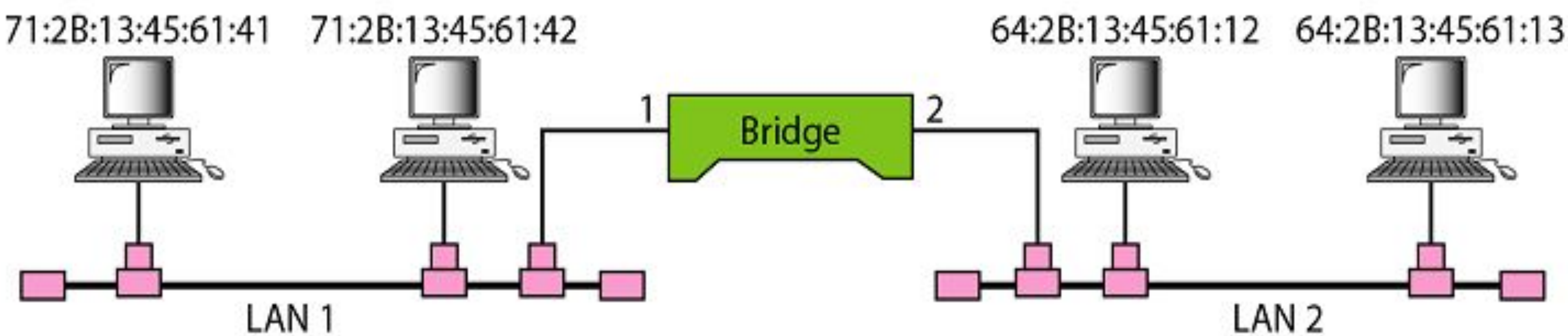
- A bridge has a **table** used in **filtering decisions**.
- It can **check the destination address** of a frame and **decide if the frame should be forwarded** or **dropped**.
- For frame to be forwarding, it specify the port.
- **Limit or filter traffic** - **keeps local traffic local** yet allow connectivity to other parts (segments).

A bridge connecting two LANs



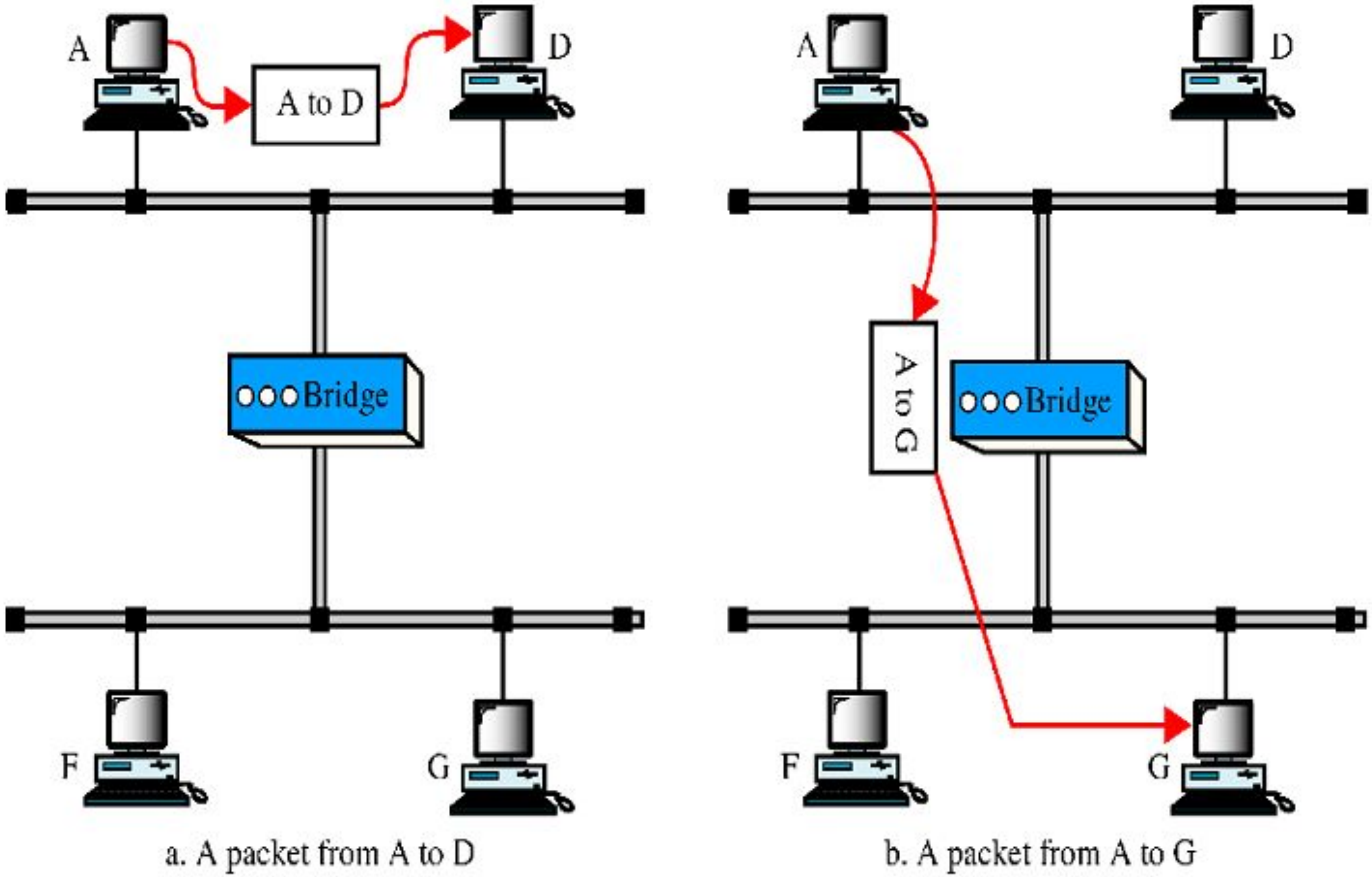
Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table



A bridge does not change the physical (MAC) addresses in a frame.

Function of Bridge



Characteristics of Bridges

- **Routing Tables**
  – Contains **one entry per station** of network.
  – Is used to determine the network of destination.
- **Filtering**
  – Packets are filtered with respect to their destination and multicast addresses.
- **Forwarding**
  – the process of passing a packet from one network to another.
- **Learning Algorithm**
  – the process by which the bridge learns how to reach stations on the internetwork.

Types of Bridges

- **Transparent Bridge**
  - Also called **learning bridges**
  - Build a table of MAC addresses as frames arrive.
  - **Ethernet networks use transparent bridge**
  - Duties are : **Filtering frames**, **forwarding and blocking**
- **Source Routing Bridge**
  - Used in **Token Ring networks**
  - Frame contains not only the **source and destination address** but also the **bridge addresses**.

Advantages And Disadvantages

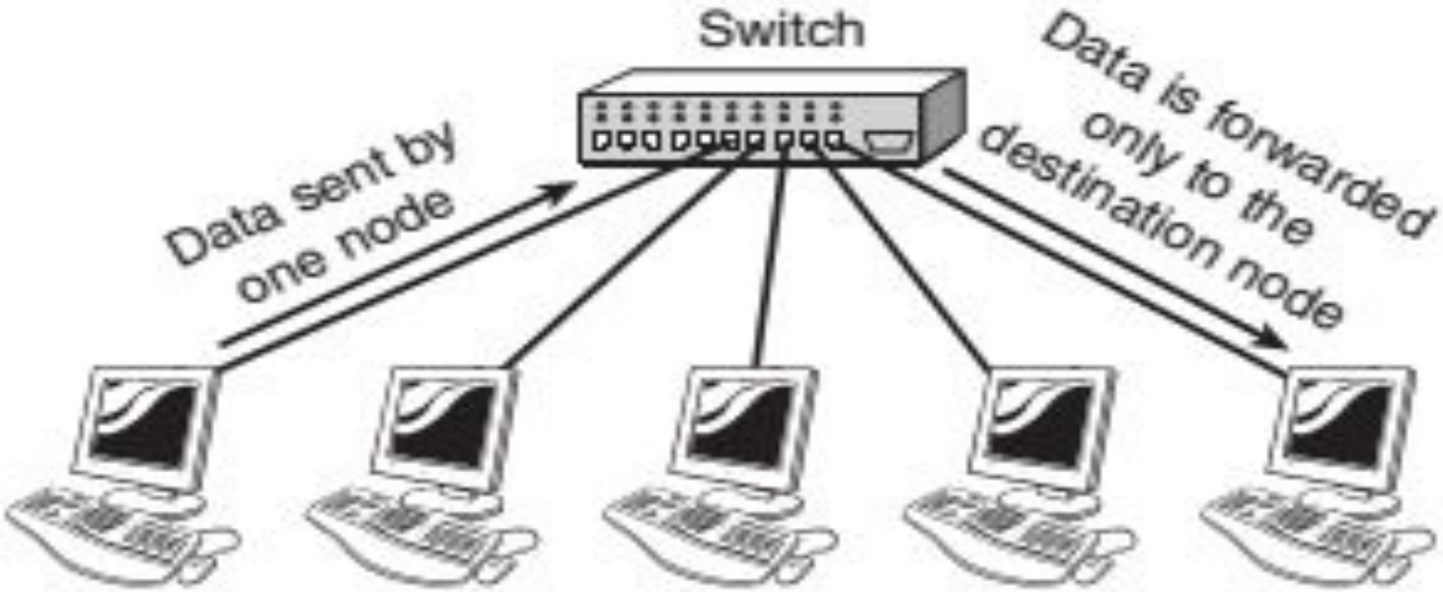
- **Advantages of using a bridge**
  - Extend physical network
  - Reduce network traffic with minor segmentation
  - Reduce collisions
  - Connect different architecture
- **Disadvantages of using bridges**
  - **Slower** than repeaters due to filtering
  - **Do not filter broadcasts**
  - **More expensive** than repeaters

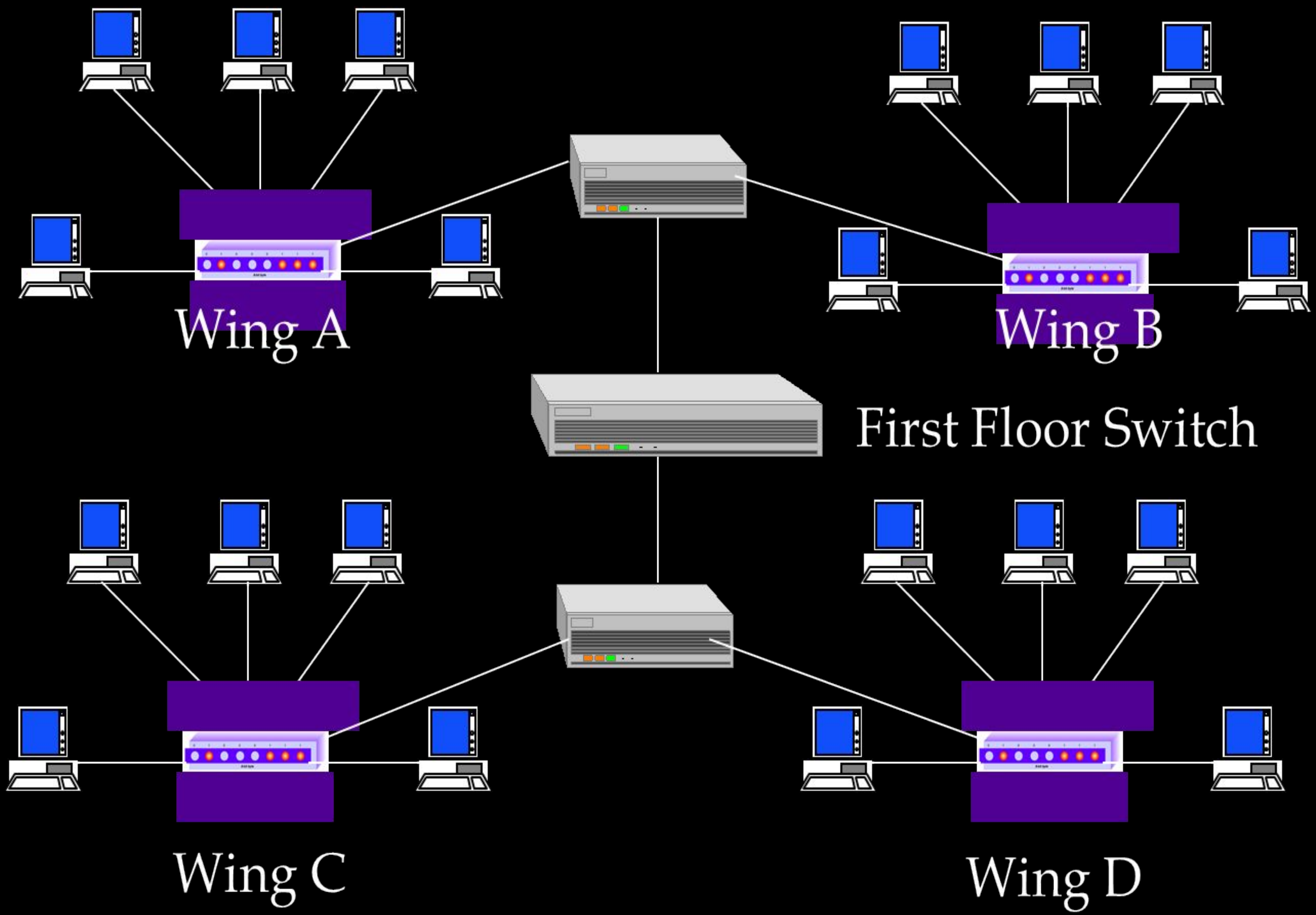
Differences Between Bridges and Repeaters

	Repeaters	*Bridges*
OSI layer	Physical layer	Data link layer
Data regeneration	Regenerate data at the signal level	Regenerate data at the packet level
Reduce network traffic	No	Yes

Switch

- Switches are a **special type of hub.**
- Switch is used as a **central device** to connect different nodes in star topology.
- Switch is an **intelligent device**.
- When switch receives a data packet it reads **destination address** stored in the packet.
- and send it to **only that node whose destination address matches** with the address contained in the data packet.
- Switches operate at **the data link layer of the OSI Model.**


Switch
Data sent by one node
Data is forwarded only to the destination node


Wing A
Wing B
First Floor Switch
Wing C
Wing D

Advantages of Switches

- **Switches divide a network into several separate channels**
- **Reduce the possibility of collision**
- **Each channel has its own network capacity**
- **Since isolated, hence secure**
    - **Data will only go to the destination, but not others**
- Enable all ports to work at the same time
- can convert protocols
- configurable
- high speed

Disadvantages

- understand **only data link layer protocols** and addresses
- much more **expensive** than previous options.
- **higher maintenance**

Types of Switches
Cut Through Switch

- In a cut-through switching environment, the packet begins to be forwarded **as soon as it is received**.
- This method is very fast.
- **Faster than store and forward** because doesn't perform error checking on frames

- **Forwards bad frames**
  - Performance penalty because bad frames can't be used and needs to be resent.

Method of Switching - Cut Through Mode

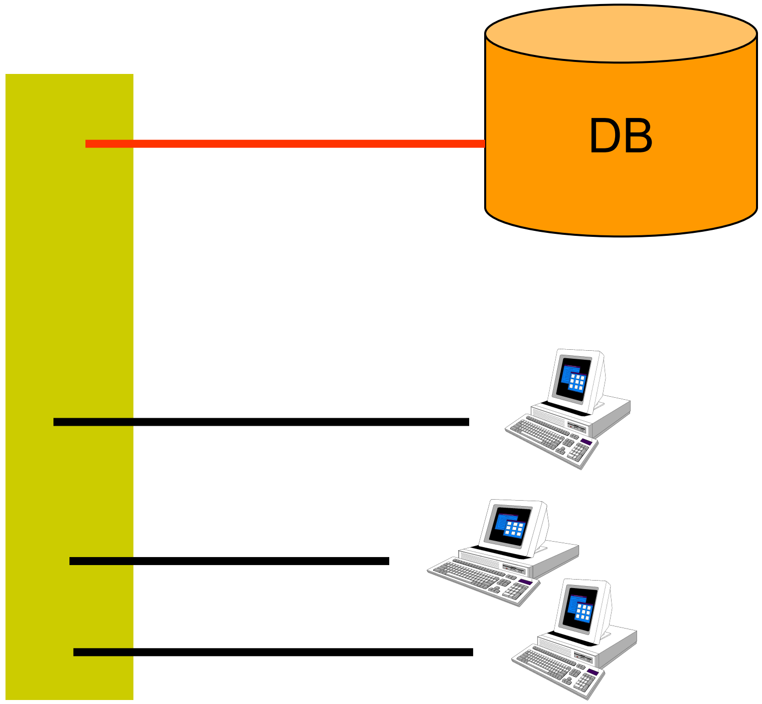
Preamble		Des. Add	Sour. Add	Length	Data	FCS
7 Bytes	1 Byte	2/6 Bytes	2/6 Bytes	2 Bytes	46 - 1500 Bytes	4 Bytes

- Read the first 14 bytes of each packet, then transmit
- Much faster
- Cannot detect corrupt packets

Store and Forward Switches

- Unlike cut-through, in a store-and-forward switching environment, the **entire packet is received** and **error checked before being forwarded**.
- The upside of this method is that errors are not propagated through the network.
- **Slower than other types** of switches
  - because it holds each frame until it is completely received and check for errors before forwarding

Method of Switching - Store and Forward

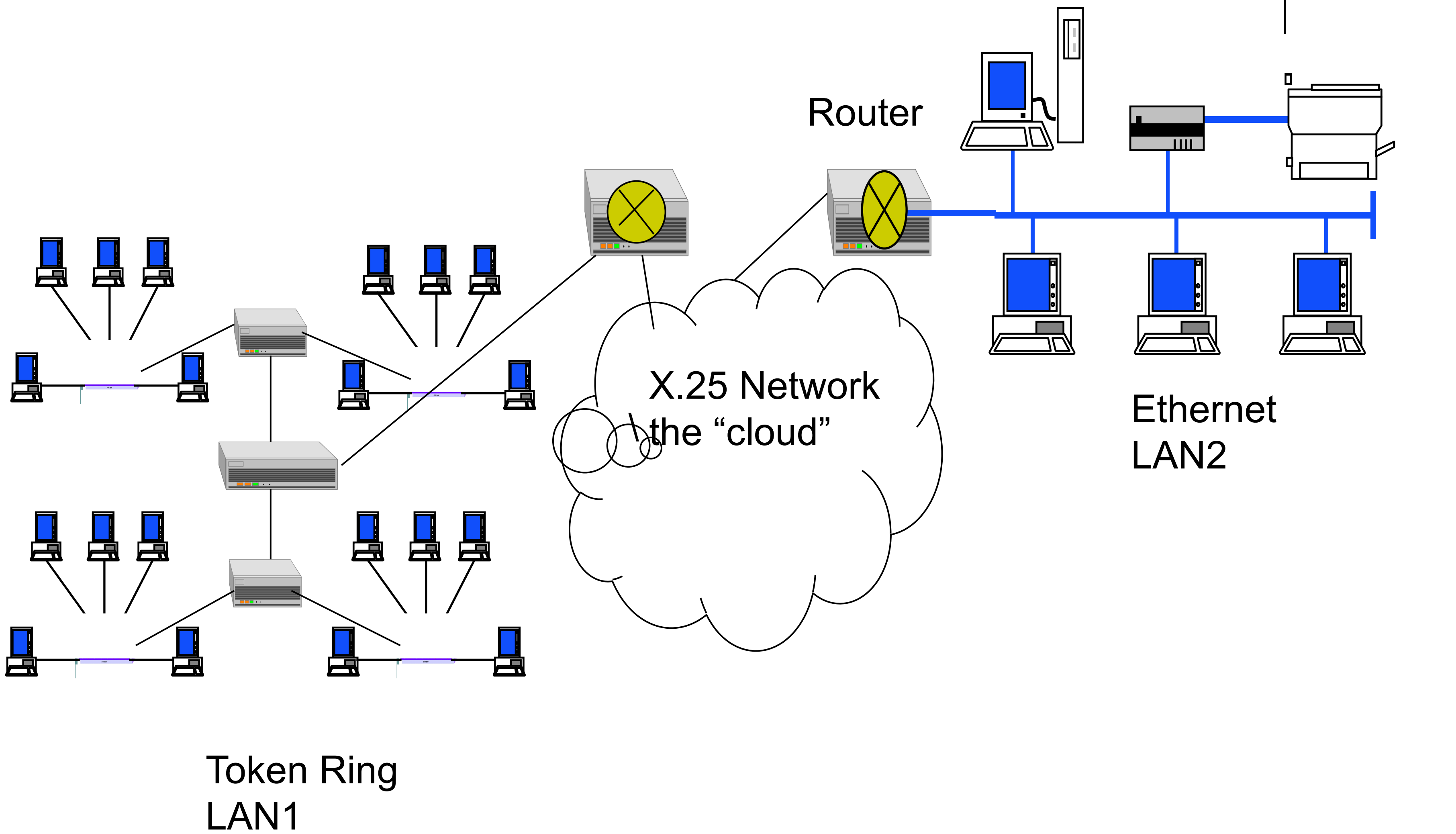


Fragment free cut through switch

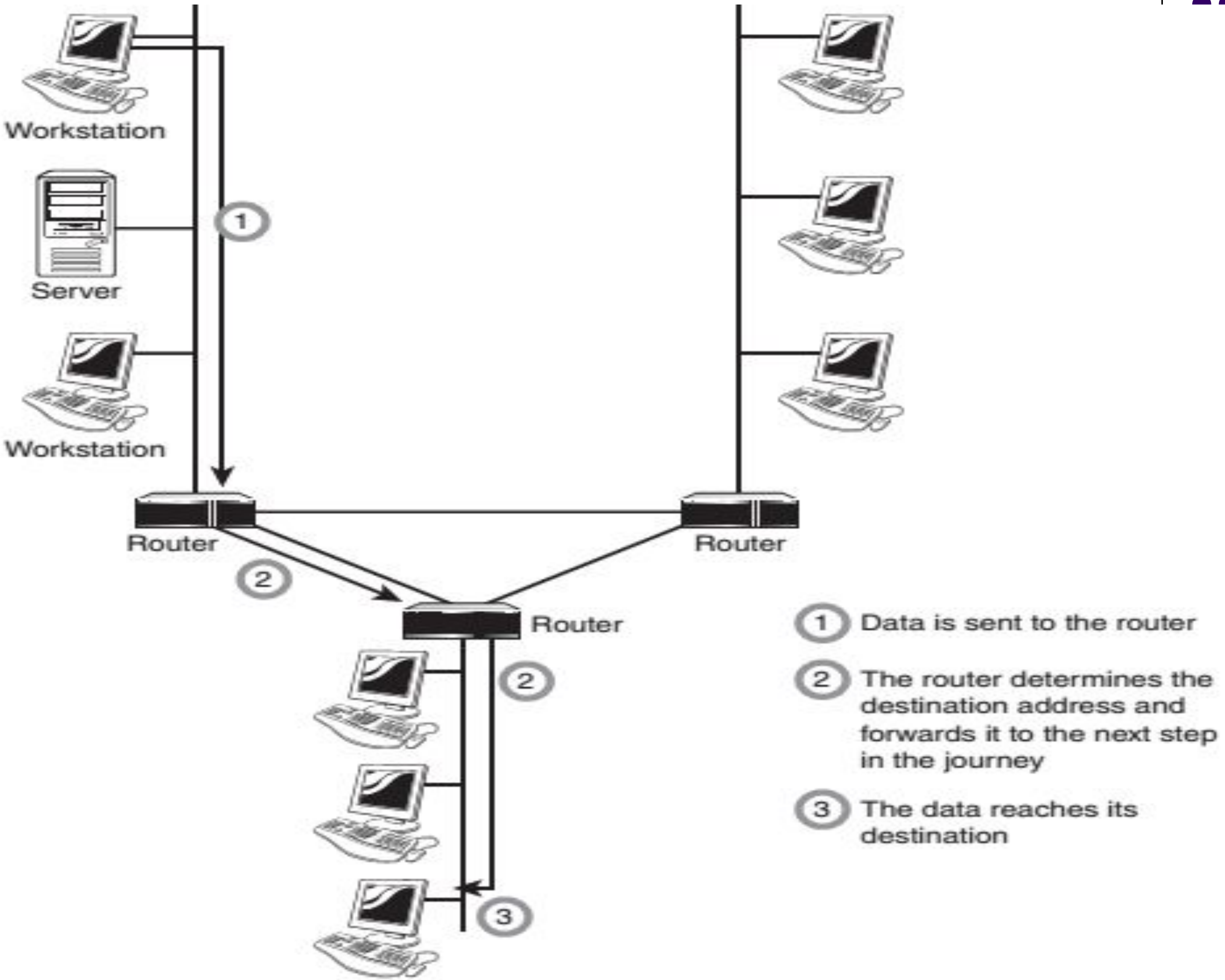
- Combines **speed of cut through switch** with **error checking functionality**
  - Forwards all frames initially,
  - but determines that if a particular port is receiving too many bad frames,
  - it reconfigures the port to store and forward mode.

Routers

- Routers are used to **create larger networks** by **joining two or more network segments**.
- A router derives its name from the fact that **it can route data** it receives from one network onto another.
- When a router receives a packet of data, it determines the **destination address**.
- And then, it looks in its **routing table** to determine, how to reach the destination.
- Routers **work at the OSI layer 3 (network layer)**


Router
X.25 Network
the "cloud"
Ethernet
LAN2
Token Ring
LAN1

- Router consist of **hardware and software**.
  - **Hardware** - physical interfaces to the various networks in the internetwork.
  - **Software**- is OS and routing protocols management software.
- Router use **logical and physical addressing** to connect two or more logically separate networks called subnets.
- Each of the subnet is given a **logical address**.
- Data is grouped into **packets**. Each packet has physical device address and logical network address.


Workstation
Server
Workstation
1
Router
Router
2
Router
2
3
1 Data is sent to the router
2 The router determines the destination address and forwards it to the next step in the journey
3 The data reaches its destination

Advantages of Routers

- Advantages of routers:
  - Routers can connect **different types of network.**
  - Routes can choose the **best path** across the network.
  - Routers **reduce network traffic** because they do not retransmit network broadcast traffic.

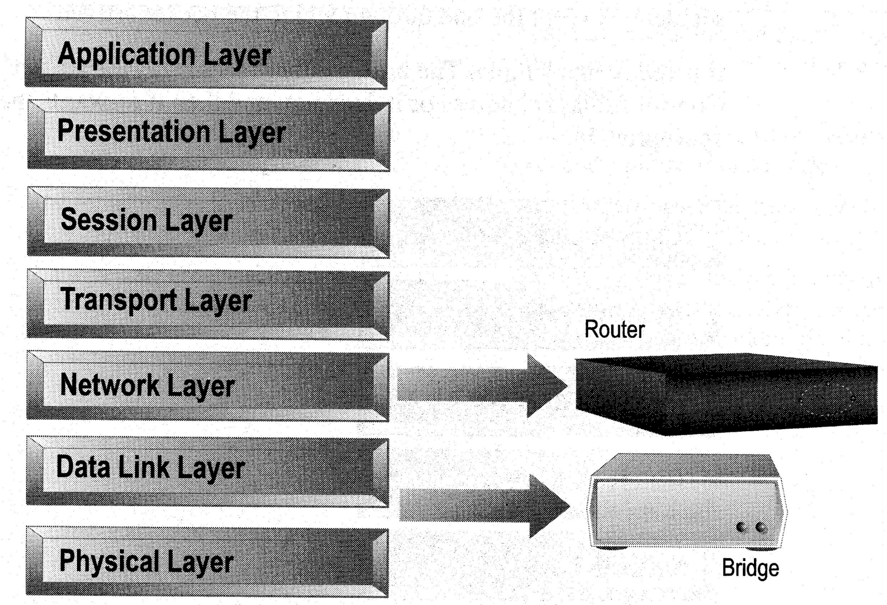
Disadvantages of Routers

- Disadvantages of routers:
  - Routers are **more expensive** than bridges or repeaters.
  - **Routers are slower** than **bridges or switches** because they must analyze a data transmission from the Physical through the Network layer

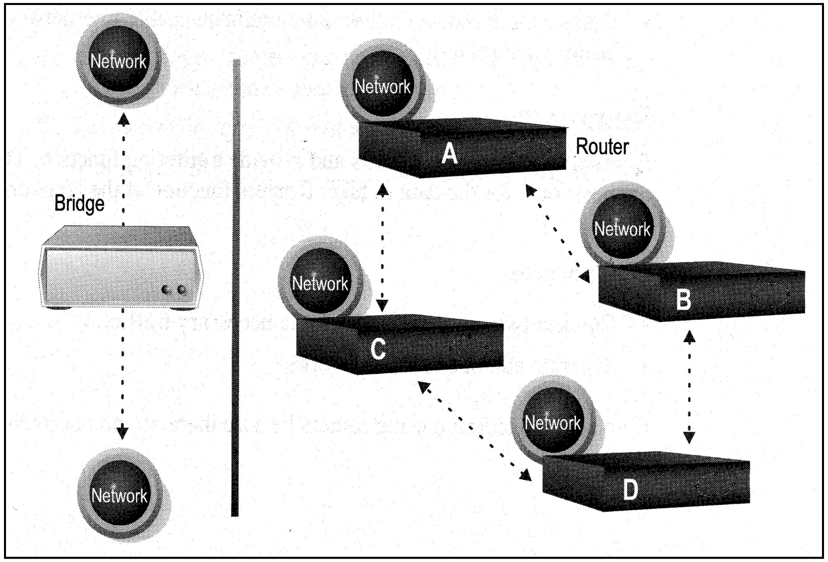
Static and Dynamic Routers

Static Routers	*Dynamic Routers*
Manual configuration of routes	Manual configuration of the first route. Automatic discovery of new routes
Always use the same route	Can select the best route
More secure	Need manual configuration to improve security
static routing is suited to only the smaller network	suited to larger network environment

Distinguishing Between Bridges and Routers



- **Bridges** forward everything they don't recognize
- **Routers** select the best path



- **Routers** are layer 3 devices which recognize network address
- **Bridges** are layer 2 devices which look at the MAC sublayer node address

Brouters

- Device that functions as:
  - A bridge for non-routable protocols
  - A router for routable protocols
- Operates at both the Data Link and Network layers.

Gateways

- Any device that **translates one data format to another** is called a ***gateway***.
- The key point about a gateway is that **only the data format is translated**, not the data itself.
- Gateways
  - Combination of hardware and software

Gateways

- Gateways are complex machines that interfaces between two or more ***dissimilar*** networks
- Operates at the **network layer (3) or higher layers** (4-7 **session, presentation, and application layers**)
- Forwards only those messages that need to go out

Gateways

- **Categories:**
  - **E-mail gateway** - translates messages from one type of e-mail system to another.
  - **Internet gateway** - allows and manages access between LANs and the Internet.
  - **LAN gateway** - allows segments of a LAN running **different protocols**, **network access methods**, or **transmission types** to communicate with each other.
  - **Voice/data gateway** - allows a data network to issue data signals over a voice network.
  - **Wireless gateway** - integrates a wireline network with a wireless network.

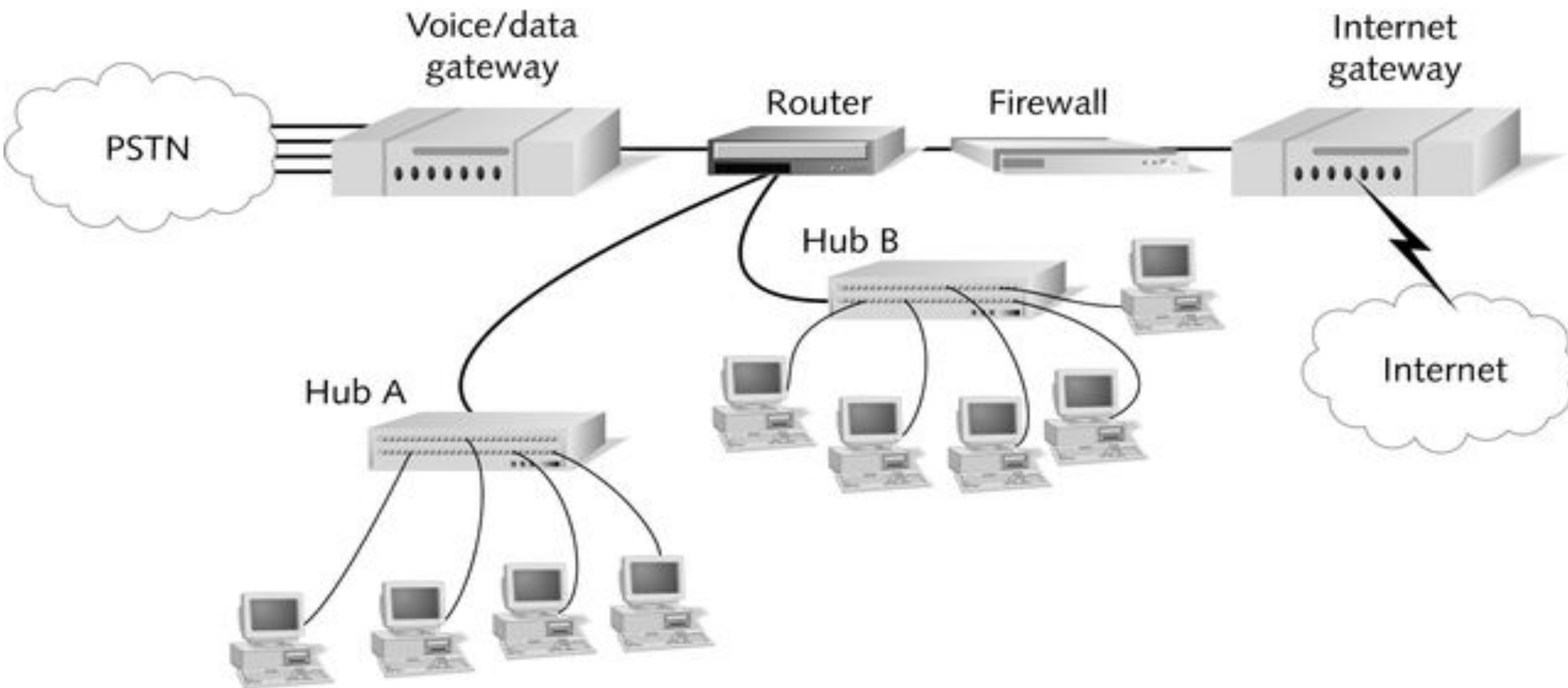


Figure 11-13 Gateways on a network

State the situations under which gateways are necessary in the network. (Necessary2Marks. example2Marks)

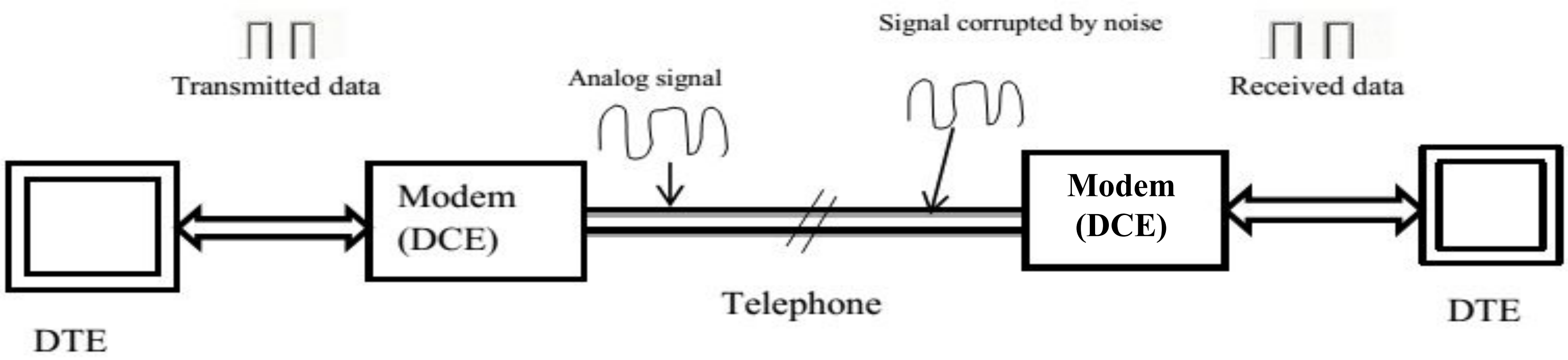
- Gateway operates at **all 7 layers** of the OSI model.
- **Situation where gateways are necessary** for different n/w like Ethernet, Token Ring, and FDDI etc. They can communicate if they are using same protocol or different protocol.
- Eg: if n/w A is a Token Ring network using TCP/IP & network B is a Novell Network, a gateway can relay frames between two.
- This means that a gateways **translate between different protocols**.
- In certain situations the only changes required are to the frame header.
- In other cases, the gateway must take care of **different frame sizes, data rates, format, acknowledgement** schemes, and priority schemes tec.

Modem

- A ***modem***, short for **Modulator/demodulator**, is a device that **converts the digital signals** generated by a computer **into analog signals** that can travel over conventional **phone lines**.

- The modem at the receiving end converts the **signal back into a format the computer can understand**.

- Allow computers to communicate over a **telephone line**.



- Modems are available as **Internal or external** modem.

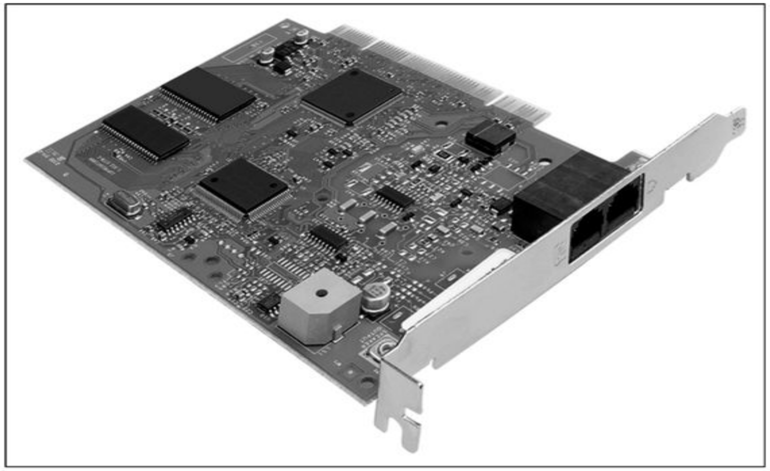


Figure 11-15 An adapter card modem

Figure 11-16 A serial port modem

Figure 11-17 A PC Card modem

Network Interface Card(NIC)

- NIC is a small card inserted or plugged on the motherboard of the host.
- It has a small CPU, memory and a limited instruction set required for the network related functions.
- Each NIC has **a unique hardware address** or **physical address** to identify the host uniquely, which ensures that its unique all over world.

The functions include

- It accepts instructions from host to transfer data to cable and receive data from cable.
- It checks the status of the bus and sends the data bit by bit once the bus is idle.
- It inserts the **CRC** in the header of the frame while transmitting.
- While accepting the data, NIC compares the destination address in the frame with its own hardware address;
- If matches then only it is accepted otherwise rejected.
- Validating the input frame by checking its **CRC** to ensure that the data is error free.

Network Software

- **Device Driver**:
  - **Computer Program** that operates or controls a particular type of device connected to computer.
  - The driver **sends commands** to the device.
  - Drivers are **hardware dependent** and **OS specific**.
  - Eg. Printer, Video adapters, network cards, Sound cards, scanners , etc.

NIC Device Drivers

- NIC is adapter that plugs into computer and allows to send and receive signals on a network.
- NIC driver software is installed on a computer.
- Drivers are manufacturer specific.
  - Plug and play drivers.
  - Manual installation.
  - Add remove hardware wizard.
- Set the IP Addresses.

Client-Server Software

- **TELNET :**
- TELNET is abbreviation for **Terminal Network**.
- It is standard TCP/IP protocol for virtual terminal services proposed by ISO.
- TELNET enables establishment of connection to a **remote system,** in such a way that a local terminal appears to be terminal at remote system.
- TELNET is general purpose client server application program.

FTP-File Transfer Protocol

- FTP is a standard mechanism provided by the Internet for **copying a file from one host to the other**.
- Some of the problem in transferring files from one system to the other are as follows:
  - Two systems may use **different file name conventions**.
  - Two systems may represent **text data in different types**.
  - The **directory structure** of the two systems may be different.
- FTP provides a simple solution to all these problems.

- FTP establishes **two connections** between the **client and server.**
  - One is for **data transfer** and
  - the other is for the **control information**.
- The control connection uses simple rules of communication. Only one line of command or a line of response is transferred at a time.
- But the data connection uses more complex rules due to the variety of data types being transferred.
- FTP uses **port 21** for the **control connection** and **port 20** for the **data connection**.